

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 1st Semester B.Sc. in CSE Final Examination, 2023
CSE-1101: Fundamentals of Computers and Computing

Total Marks: 70

Time: 3 Hours

Answer any 5 (Five) of the following Questions

1.
 - a) Define Computer. With an appropriate diagram explain the organization of a computer. 5
 - b) Explain why modern computers contain both Dynamic RAM (DRAM) and Static RAM (SRAM). 4
 - c) Write short note on alphanumeric codes. 3
 - d) Define (i) Port (ii) Bus. 2
2.
 - a) Define Software. Distinguish between system software and application software. 5
 - b) Write down the characteristics of machine, assembly and high level languages. 5
 - c) Differentiate among PROM, EPROM and EEPROM. 4
3.
 - a) Define the terms: HTTP, URL, protocol and DNS Server. 4
 - b) What are meant by 127.0.0.1 address and local-host? 2
 - c) What is network topology? Write down the characteristics of different network topologies. 5
 - d) What do you understand by (i) Intranet (ii) Extranet? 3
4.
 - a) Explain the process management functions of an operating system. 5
 - b) How does cache memory improve the performance of the system? 4
 - c) Explain how read/write operation is carried out in a magnetic disk. 5
5.
 - a) Write short notes on (i) LCD monitor, (ii) scanner and (iii) printer. 6
 - b) Convert the number $(642.75)_8$ to decimal, binary and hexadecimal. 3
 - c) Perform the following operations: 3
 - (i) Add 1011001 and 11010 (ii) Subtract 1001 from 1101 (iii) Divide 1110 by 0111
 - d) What is co-processor? 2
6.
 - a) Discuss different data types used in C with proper example. 4
 - b) Briefly describe 'nested if' condition in C with appropriate code example. 5
 - c) Write a program to check whether an alphabet is a vowel or consonant. 5
7.
 - a) Define Variable. What is the difference between local variable and global variable? 4
 - b) What type of function is main() in C? 3
 - c) What is header file in C? Why do we use it? 3
 - d) Write a C program to find the roots of a quadratic equation. 4